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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting makes it harder to search for items
- b. Sorting is rarely used in programming
- c. Sorting decreases the efficiency of selection operations
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It increases the number of comparisons needed
- c. It has no effect on the search operation
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It does not require recursion
- c. It is less efficient than bubble sort
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **4**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Backtracking
- b. Dynamic programming
- c. Divide and conquer
- d. Greedy method

The correct answer is:
Divide and conquer

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Merge Sort
- c. Bubble Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Insertion Sort
- b. Merge Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Makes data harder to manage
c. Decreases the efficiency of algorithms
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **10**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It uses the heap data structure
c. It bubbles up the largest element in each pass
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sorted()
c. sort()
d. order()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It has no effect on the analysis process
c. It makes the analysis slower
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **13**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By sorting data in a single pass
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Finding a specific element in a list
c. Removing duplicates from a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **15**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the largest element to its correct position
c. Bubbling up the smallest element
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position